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Engine-Oil Dipstick Having a Magnet

Abstract

An engine-oil dipstick having a magnet having a cap assembly that is configured to removably attach to a dipstick entry orifice of a combustion-engine sump; a dipstick element having a first end and a second end; the first end of the dipstick element attached to the cap assembly; a magnet that generates a magnetic field, the magnet having a first half and a second half; the second half of the magnet being closer to the second end of the dipstick than the first half of the magnet; the second half of the magnet being substantially encapsulated within the dipstick element.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This patent application claims priority to pending U.S. provisional patent application 63/556,859 having a filing date of Feb. 22, 2024. The subject matter of U.S. provisional patent application 63/556,859 is incorporated by reference into this application.

BACKGROUND OF THE INVENTION

[0002] There remains a need for filtering or removing ferromagnetic particulates from engine oil within a combustion-engine sump.

BRIEF SUMMARY OF THE INVENTION

[0003] An engine-oil dipstick having a magnet having a cap assembly that is configured to removably attach to a dipstick entry orifice of a combustion-engine sump; a dipstick element having a first end and a second end; the first end of the dipstick element attached to the cap assembly; a magnet that generates a magnetic field, the magnet having a first half and a second half; the second half of the magnet being closer to the second end of the dipstick than the first half of the magnet; the second half of the magnet being substantially encapsulated within the dipstick element.

[0004] An engine-oil dipstick having a magnet having a cap assembly that is configured to removably attach to a dipstick entry orifice of a combustion-engine sump; a dipstick element having a first end and a second end; the first end of the dipstick element attached to the cap assembly; a magnet that generates a magnetic field, the magnet having a first half and a second half; the second half of the magnet being closer to the second end of the dipstick than the first half of the magnet; the second half of the magnet being substantially encapsulated within the dipstick element; the second half of the magnet having a second-half-of-the-magnet surface area; the portion of the second end of the dipstick that substantially encapsulates the second half of the magnet having an exterior surface; the exterior surface having an exterior-surface surface area; the exterior-surface surface area being greater than the second-half-of-the-magnet surface area.

[0005] An engine-oil dipstick having a magnet having a cap assembly that is configured to removably attach to a dipstick entry orifice of a combustion-engine sump; a dipstick element having a first end and a second end; the first end of the dipstick element attached to the cap assembly; a magnet that generates a magnetic field, the magnet having a first half and a second half, the second half of the magnet being closer to the second end of the dipstick than the first half of the magnet; the second half of the magnet being substantially encapsulated within the dipstick element; the dipstick element having a length that, upon attaching the cap assembly to a dipstick entry orifice of a combustion-engine sump, is configured to at least partially submerge the second half of the magnet that is substantially encapsulated within the dipstick element beneath a liquid-fill level within a combustion-engine sump.

[0006] An advantage of the disclosed embodiments is that at the distal end of the dipstick element, the distal end of a magnetic element is substantially encapsulated therein. Because of this substantial encapsulation, the surface area upon which ferromagnetic materials can collect is increased beyond the magnet's natural surface area.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0007] FIG. 1 shows a perspective view of a dipstick embodiment.

[0008] FIG. 2 shows an exploded perspective view of a dipstick embodiment.

[0009] FIG. 3 shows a side view of a dipstick embodiment.

[0010] FIG. 4 shows an exploded side view of a dipstick embodiment.

[0011] FIG. 5 shows a perspective view of an embodiment.

[0012] FIG. 6 shows an exploded perspective view of a dipstick embodiment.

[0013] FIG. 7 shows a schematic of an embodiment in use.

[0014] FIG. 8 shows an exemplary use of an embodiment.

DETAILED DESCRIPTION OF THE INVENTION

[0015] Embodiments are directed to engine-oil dipstick **100** having magnet **110** that generates magnetic field **118**.

[0016] As shown in the figures, engine-oil dipstick **100** includes cap assembly **120** having cap-assembly threading element **122** that is configured to mate with complimentary threading element **144** on a dipstick entry orifice **142** of a combustion-engine sump **140**.

[0017] In embodiments shown in FIGS. 1-4, cap assembly **120** is fixedly attached to first end **152** of dipstick element **150** such that dipstick element **150** is in a fixed position relative to cap assembly **120**. In these fixed-position embodiments, cap assembly **120** includes cap knob **206** and cap-assembly threading element **122**.

[0018] In other embodiments, shown in FIGS. 5-6, cap assembly **120** is rotatably attached to first end **152** of dipstick element **150** via ball **200** and socket **202** such that dipstick element **150** is movable relative to cap assembly **120**; the attached connection that is achieved by ball **200** and socket **202** allows dipstick element **150** to assume a variety of angular positions relative to cap assembly **120**. In embodiments that include ball **200** and socket **202**, and as shown in FIGS. 5-6, cap assembly **120** includes cap knob **206**, plug **204**, cap-assembly threading element **122**, and socket **202** (that mates with ball **200** that is fixedly attached to first end **152** of dipstick element **150**).

[0019] Dipstick element **150** has first end **152**, second end **154**, first half of the dipstick element **156** (that includes first end **152**), and second half of the dipstick element **158** (that includes second end **154**). As shown in FIGS. 1, 3, and 5; and because these figures show dipstick element **150** as having an opaque exterior surface, magnet **110** is shown with an arrow as being substantially encapsulated within second half of dipstick element **158** within or proximate to second end **154**. Exploded views shown in FIGS. 2, 4, and 6 show magnet **110** located within or proximate to dipstick-element second end **154**. To add even more clarity to embodiment positions of magnet **110** within or proximate to dipstick-element second end **154**, orifice **160** is configured to allow visibility of magnet **110** that is substantially encapsulated within second half of dipstick element **158**.

[0020] In embodiments in which magnet **110** is described as being substantially encapsulated within second half of dipstick element **158**, "substantially encapsulated" means that at least 90% of magnet **110** surface area is covered by second half of dipstick element **158**. In other embodiments, substantially encapsulated means that at least 80% of magnet **110** surface area is covered by second half of dipstick element **158**. In still other embodiments, substantially encapsulated means that at least 70% of magnet **110** surface area is covered by second half of dipstick element **158**.

[0021] As shown in FIG. 2, magnet **110** has magnet first half **112**, magnet second half **114**, and second-half-of-the-magnet surface area **116**. Magnet first half **112** is positioned closer to dipstick-element first end **152** than magnet second half **114**; likewise, magnet second half **114** is positioned closer to dipstick-element second end **154** than magnet first half **112**. In embodiments, at least magnet second half **114** is substantially encapsulated within second half of dipstick element **158**; in other embodiments, both magnet first half **112** and magnet second half **114** are substantially encapsulated within second half of dipstick element **158**. In embodiments, either magnet first half **112** or magnet second half **114** is visible through orifice **160**.

[0022] In embodiments in which magnet second half **114** is described as being substantially encapsulated within second half of dipstick element **158**, at least 90% of second-half-of-the-magnet surface area **116** is covered by second half of dipstick element **158**. In other embodiments, at least

80% of second-half-of-the-magnet surface area **116** is covered by second half of dipstick element **158**. In still other embodiments, at least 70% of second-half-of-the-magnet surface area **116** is covered by second half of dipstick element **158**.

[0023] As shown in the figures, dipstick-element portion **1510** substantially encapsulates magnet second half **114**. Dipstick-element portion **1510** has dipstick-element exterior surface **1512** that has a surface area. Because magnet second half **114** is encapsulated within dipstick-element portion **1510**, the surface area of dipstick-element exterior surface **1512** is necessarily greater than second-half-of-the-magnet surface area **116**. Mathematically, this can be generally understood as second-half-of-the-magnet surface area **116** $X.\sup.2$ being less than the dipstick-element surface **1512** having a surface area $[X+Y].\sup.2$ wherein both X and Y are positive integers; stated differently, $X.\sup.2 < [X+Y].\sup.2$. This increased surface area (represented as $[X+Y].\sup.2$), due to encapsulation of magnet **110**, is advantageous because it provides relatively more dipstick-element surface area upon which ferromagnetic materials **210** can be attracted to and collect than would otherwise be available if magnet **110** were not encapsulated.

[0024] Dipstick element **150** has dipstick-element length **170** that, upon attaching cap assembly **120** to a dipstick-entry orifice **142** of a combustion-engine sump **140** (via cap-assembly threading element **122** screwing into or mating with complimentary threading element **144** on a dipstick-entry orifice **142** of a combustion-engine sump **140**), is configured to at least partially submerge magnet second half **114** (that is substantially encapsulated within dipstick element **150**) beneath a liquid-fill level **146** within the combustion-engine sump **140**. Persons of ordinary skill in the art will be able to determine a useful dipstick-element length **170** without having to exercise undue experimentation.

[0025] FIGS. 7 and 8 show engine-oil dipstick **100** in use within combustion-engine sump environment **130**; FIG. 7 provides a schematic of combustion-engine sump environment **130** and FIG. 8 shows engine-oil dipstick **100** in use in a more specific combustion-engine sump environment **130** that is part of a motorcycle. FIG. 7 shows magnet **110** generating magnetic field **118** that is attracting and collecting ferromagnetic materials **210** from engine-oil fluid flow represented by a plurality of arrows (shown circulating in a generally counter-clockwise direction). Ferromagnetic materials **210** are shown as being attracted to and collecting on dipstick-element wall **1514**/dipstick-element wall exterior surface **1516** (shown in FIGS. 2, 4, 5, and 6). Dipstick-element wall **1514** has dipstick-element wall thickness **1518** that enables magnetic field **118** to pass through dipstick-element wall **1514** and thereby attract and collect ferromagnetic materials **210** on dipstick-element wall exterior surface **1516**. Very generally, useful dipstick-element wall thickness(es) **1518** can be determined by persons having ordinary skill in the art without having to exercise undue experimentation; in embodiments, dipstick-element wall thickness **1518** is kept at a minimum to allow magnetic field **118** to attract ferromagnetic materials **210** that are in position **190** that is proximate to and outside of dipstick-element wall **1514**.

[0026] In embodiments, dipstick element **150** is manufactured at least in part with a polymeric composition that can be selected by persons of ordinary skill in the art without having to exercise undue experimentation.

Claims

1. An engine-oil dipstick having a magnet comprising: a cap assembly that is configured to removably attach to a dipstick entry orifice of a combustion-engine sump; a dipstick element having a first end and a second end; the first end of the dipstick element attached to the cap assembly; a magnet that generates a magnetic field, the magnet having a first half and a second half; the second half of the magnet being closer to the second end of the dipstick than the first half of the magnet; and the second half of the magnet being substantially encapsulated within the dipstick element.

2. The engine-oil dipstick having a magnet of claim 1, further comprising: the second half of the magnet having a second-half-of-the-magnet surface area; the portion of the second end of the dipstick that substantially encapsulates the second half of the magnet having an exterior surface; the exterior surface having an exterior-surface surface area; and the exterior-surface surface area being greater than the second-half-of-the-magnet surface area.
 3. The engine-oil dipstick having a magnet of claim 1, further comprising: the first half of the magnet being substantially encapsulated within the dipstick element having a first end and a second end.
 4. The engine-oil dipstick having a magnet of claim 1, further comprising: the dipstick element having a dipstick-element first half that includes the first end and a dipstick-element second half that includes the second end; and an orifice in the dipstick-element second half configured to allow visibility of the magnet.
 5. The engine-oil dipstick having a magnet of claim 1, further comprising: the dipstick element having a length that, upon attaching the cap assembly to a dipstick entry orifice of a combustion-engine sump, is configured to at least partially submerge the second half of the magnet that is substantially encapsulated within the dipstick element beneath a liquid-fill level within a combustion-engine sump.
 6. The engine-oil dipstick having a magnet of claim 1, further comprising: the dipstick element having a dipstick wall; the dipstick wall having a dipstick-wall exterior surface; and the dipstick wall having a dipstick-wall thickness that enables the magnetic field to attract and collect ferromagnetic materials on the dipstick-wall exterior surface.
 7. The engine-oil dipstick having a magnet of claim 1, further comprising: the cap assembly having a cap-assembly threading element configured to mate with a complimentary threading element on a dipstick entry orifice of a combustion-engine sump.
 8. The engine-oil dipstick having a magnet of claim 6, wherein the dipstick wall is polymeric.
 9. An engine-oil dipstick having a magnet comprising: a cap assembly that is configured to removably attach to a dipstick entry orifice of a combustion-engine sump; a dipstick element having a first end and a second end; the first end of the dipstick element attached to the cap assembly; a magnet that generates a magnetic field, the magnet having a first half and a second half; the second half of the magnet being closer to the second end of the dipstick than the first half of the magnet; the second half of the magnet being substantially encapsulated within the dipstick element; the second half of the magnet having a second-half-of-the-magnet surface area; the portion of the second end of the dipstick that substantially encapsulates the second half of the magnet having an exterior surface; the exterior surface having an exterior-surface surface area; and the exterior-surface surface area being greater than the second-half-of-the-magnet surface area.
 10. An engine-oil dipstick having a magnet comprising: a cap assembly that is configured to removably attach to a dipstick entry orifice of a combustion-engine sump; a dipstick element having a first end and a second end; the first end of the dipstick element attached to the cap assembly; a magnet that generates a magnetic field, the magnet having a first half and a second half; the second half of the magnet being closer to the second end of the dipstick than the first half of the magnet; the second half of the magnet being substantially encapsulated within the dipstick element; and the dipstick element having a length that, upon attaching the cap assembly to a dipstick entry orifice of a combustion-engine sump, is configured to at least partially submerge the second half of the magnet that is substantially encapsulated within the dipstick element beneath a liquid-fill level within a combustion-engine sump.
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